08F — Deep-Dive Feature Audit: Outside-Sect World & Ecology (Part 6 of 6 — Grand Finale)

Master Holistic Overview of the Entire Outside-Sect World & Ecology: Quantum Bilocation Deployment Bugs, Visiting-Merchant `resourcePool` Leakage, 200-Year Continental Dynamism & Master `P0` $\to$ `P3` Roadmap

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Role of Document `08F`**: **Master Holistic Synthesis** concluding the 6-part Outside-Sect World & Ecology Deep Audit (`08A–08F`), examining macro-economic conservation, spatial/deployment invariants, and multi-century world evolution across `expedition.js`, `explore.js`, `external_market.js`, `wildlife.js`, `wild_resources.js`, `fishing.js`, `visitors_clues.js`, `sect.js`, and `sim.js`.
- **Immutable Project Rules**:
  1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
  2. **Preserve exact 40 `<script>` tags** and `<= 3.5 ms/tick` 80-NPC performance budget.
  3. **Single Physical Presence & Closed-Loop World Conservation**.

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1. `[WORLD-SYS-01]` Critical Quantum-Bilocation Bug: One Disciple Can Garrison Multiple Distant Frontier Outposts, Guard Leylines, Sit in Closed-Door Seclusion, AND Farm Rice Simultaneously!

While `v2.4.0` patched `OUT-10` solely for `EXPED.assignNodeGarrison`, **four newer subsystems lack a unified physical deployment lock** (proven directly in `test/v400_living_frontier_and_sect_charter_suite.js:60–65`!):

```javascript
// v400 lines 60 & 65:
```

```
const r1 = SECT.OUTPOSTS_LOGISTICS.establishOutpost(W, 'MINE_CELESTIAL_STONE',
'CAMP_REST', [W.npcs[1].id]);
const r3 = SECT.OUTPOSTS_LOGISTICS.establishOutpost(W, 'VALLEY_MILLENNIAL_HERB',
'CAMP_SUPPLY', [W.npcs[1].id]);
```

1. **Multi-Outpost Bilocation (`SECT.OUTPOSTS_LOGISTICS.establishOutpost`)**:
    - One single NPC (`W.npcs[1].id`) can be stationed simultaneously at `MINE_CELESTIAL_STONE` (Northern Peaks) and `VALLEY_MILLENNIAL_HERB` (Southern Waters) **while remaining `n.away === false` inside the sect compound and available for expeditions**!
2. **Closed-Door Seclusion & Protector Bilocation (`CULT.enterClosedDoorSeclusion`, `v330` Phase 1)**:
    - Entering 10-day *"Closed-Door Bottleneck Seclusion (ปิดด่านทะลวงคอขวด)"* sets `disc.closedDoorSeclusion.active = true` and `protectorId = prot.id`, **yet `SIM.tick` / `AI.step` / `EXPED.depart` never check `closedDoorSeclusion.active`**—so the "sealed" cultivator walks out to mine ore, cook lunch, or join a caravan while accruing seclusion insight!
3. **Leyline Guard & Exchange Disciple Bilocation (`LEYLINE_GRID.assignAnchorGuard` & `dispatchExchangeDisciple`)**:
    - The same `guardNpcId` can be assigned to all 3 Leyline Anchors simultaneously while `away` on an expedition, and an exchange disciple studying at an allied sect stays `away === false` at home!

Clean Unified Deployment Registry (`SIM.getDeploymentState(W, n)`):

Introduce an O(1) field `n.deployment = { kind, targetId, sinceDay, untilTick }` (while keeping legacy `v400` synthetic single-line test calls backward-compatible if `opts.allowReassign` reassigns cleanly from the previous outpost!):

- An NPC with an active exterior deployment (`EXPEDITION`, `NODE_GARRISON`, `OUTPOST_GARRISON`, `FOREIGN_EXCHANGE`) has `n.away = true` and cannot be double-booked onto another outpost/expedition without first relieving their prior post!
- An NPC in `CLOSED_DOOR_SECLUSION` (or `DETAINED_JAIL`) stays inside their seclusion chamber / cell (`AI.step` skips ordinary chores; `protectorId` prioritizes guarding nearby!).

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2. [WORLD-SYS-02] Long-Horizon W.world.resourcePool Collapse & Visiting-Merchant Spirit-Stone Black Hole (external_market.js:186, 555 & ai.js:A.mine)!

Tracing Spirit-Stone conservation over **100–300 in-game years** exposes two unclosed macro faucets/sinks that destabilize long saves:

1. **Visiting Merchants Conjure Floating Stones on Arrival & Destroy Paid Sect Stones on Departure (`js/external_market.js:186 & 555`)!**:
    - On arrival (`EXT.openVisit`, `line 186`), each merchant spawns `280..3,000` `visitor.stones` out of thin air (`not drawn from W.world.resourcePool.stones`).
    - During their stay, our Sect Treasury (`EXT.tradeSect`) and our Disciples (`EXT.tradeMember`) pay **thousands of Spirit Stones** into `visitor.stones` to buy manuals, herbs, and repair stones.
    - On departure (`EXT.daily`, `line 555`: `W.world.visitors = W.world.visitors.filter(v => ... v.untilDay >= W.day)`), **the departing visitor object—along with 100% of `visitor.stones` paid by our sect—is deleted into thin air instead of returning to `W.world.resourcePool.stones`!**
    - **30-Year Cascade**: If the sect spends stones primarily with Visiting Merchants while drawing bounties/harvests from `W.world.resourcePool.stones` (`initial 50,000`), **`W.world.resourcePool.stones` drains irreversibly to `0`**, freezing all Closed-Loop world rewards forever!
    - **Exact Fix**: Draw `visitor.stones` from `W.world.resourcePool.stones` on `EXT.openVisit`, and **return `visitor.stones` back into `W.world.resourcePool.stones`** right before removing expired merchants in `EXT.daily`!
2. **Unbounded Mine Stone Faucet (`A.mine.done` in `js/ai.js`) & Natural Earth-Vein Equilibrium**:
    - `A.mine.done` currently mints `5..18` stones even when `0` ore is mined without debiting `W.world.resourcePool.stones`! Draw mined Spirit Stones from `W.world.resourcePool.stones` (with a bounded seasonal Celestial/Leyline natural regeneration target keeping total world circulation healthy around its macro equilibrium band).

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3. [WORLD-SYS-03] 200-Year Continental Evolution: Living Rival Dynasties, Rival-vs-Rival Node Wars, & HEAVENLY_DAO_DIRECTOR Scaling!

To fulfill the vision of a **self-sustaining, autonomous Wuxia world** where decades and centuries produce epic history without player scripting:

1. **Generational Turnover & 43-Path Diversity in Rival Sects**:
   - Let aging Patriarchs (`figures.master`) of the `19` rival sects face real lifespans and breakthroughs (`advancing in stage or perishing -> triggering EXPED.resolveSectSuccession organically`, occasionally sending defeated royal heirs fleeing to our sect or neighboring sects as in `v229`!).
   - Assign secondary/branch paths from the **31 Grand Dao Paths** across rival sects and newly formed `SPLINTER` / defector sects (capping active `allSects + splinterSects` cleanly at `24` sects by merging conquered/extinct minor sects).
2. **Dynamic Continental Hegemony & `HEAVENLY_DAO_DIRECTOR`**:
   - Allow rival sects to contest the 6 `RESOURCE_NODES`, bid on Midnight Black Market lots, and enter `OPEN` Continental Secret Realms using their own `rivalSect.treasury` and active champions!
   - Calibrate `HEAVENLY_DAO_DIRECTOR` (`SIM.monthly`) against the **top-5 combat power distribution across all sects**, awakening dormant Ancient Demon Sects or Immortal Alliances organically when any single sect approaches continental hegemony!

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4. Master Prioritized Execution Roadmap for GPT & Gemini (`P0` $\to$ `P3` Across `08A–08F`)

Phase	Priority	Target Files	Exact Deliverables Across `08A–08F`
Phase 1	**P0 — Dead Ticks, Starvation Bug & Locked Content**	`sim.js`, `ai.js`, `fishing.js`, `explore.js`	1) Move `CULT.menagerieTick(W)` out of pre-`W.sect` `SIM.create` (`sim.js:186`) into `SIM.dailyUpdate`

Phase	Priority	Target Files	Exact Deliverables Across 08A–08F
			so the Menagerie actually runs (08D). 2) Fix A.eat.can & A.eat so cooked Spirit-Fish Dishes prevent starvation when meal === 0, don't double-consume 'meal', wire studyMult, and stamp listedAtDay on Underwater Chest gear (08E). 3) Unlock Nine-Dao Pagoda Floors 10–43 (remove { maxFloor: 9 } at explore.js:742) and route nature/beast/music paths to the 4th Sanctum SANCTUM_MYRIAD_BEAST_TOTEM (explore.js:738) (08C).
Phase 2	**P1 — Unfixed OUT-* Exploits & Closed-Loop World Leaks**	explore.js, expedition.js, external_market.js, ui.js	1) Fix OUT-14 (emergencyTeleportToSect cancels/removes member from W.world.expeds) (08B) & OUT-05 (dispatchExploration & solveElementalPu

Phase	Priority	Target Files	Exact Deliverables Across `08A–08F`
			`zzleGate` cap stone payout by `W.world.resourcePool.stones`) (`08C`). 2) Recycle `visitor.stones` $\leftrightarrow$ `W.world.resourcePool.stones` on `EXT.openVisit` / departure (`08F`), fix `N-06.2` caravan manual sink, and exclude internal death-inheritance/confiscations from external `stoneIncome` in `computePeriodDigest` + fix pre-increment `c.stage` storyline thread ID (`08B`). 3) Fix `OUT-07` (`remove { free: true }` on `HERMIT_DOCTOR` in `ui.js`), `OUT-12` (split daily vs seasonal node harvest), `OUT-15` (`claimResourceNode` owner/receipt guard), `OUT-19` (`day=0` tribute), and `OUT-20` (duplicate

Phase	Priority	Target Files	Exact Deliverables Across `08A–08F`
			recruit guard) (`08A/08E`).
Phase 3	**P2 — Bilocation Lock, Missing Clues & Biome/Typo Fixes**	`sect.js`, `visitors_clues.js`, `data_beasts.js`, `wild_resources.js`, `wildlife.js`	1) Enforce unified `SIM.getDeploymentState` across Outposts, Resource Nodes, Leyline Guards, Closed-Door Seclusion, and Expeditions (`08F`). 2) Author the `6` missing `v2.0.0` Legendary Weapon Clue Templates (`27/27` reachable!) + `LEG_V230_*` clues, and fix `simulateGuardianEncounter` `'SWORD'` hardcode (`08E`). 3) Fix `BEAST_ABYSSAL_DRAGON` (`"urban" -> "water"`), `LEVIATHAN_TURTLE`, `THUNDER_DRAKE_LIZARD`, `!MAP.blocked` check in `WILD.tick`, and `'healer'`/`'herbal'` post IDs in `wild_resources.js` (`08D`).

Phase	Priority	Target Files	Exact Deliverables Across `08A–08F`
			4) Fix `'social.diplomacy' in {}` typo (`expedition.js:44`), require real rations/attrition on `departSectSiege` (`OUT-17`), real party on `REINFORCE` (`OUT-08`), combat on late anomalies (`OUT-16`), and staged `'OPEN'` Secret Realm races (`OUT-18`) (`08A/08C`).
Phase 4	**P3 — Full Emergent Jianghu & Ecological Autonomy**	`expedition.js`, `sect.js`, `cultivation.js`, `ai.js`, `visitors_clues.js`	1) Enable autonomous `reservePets` inheritance/adoption, `'acquire_spirit_beast'` goals, `BEAST_SYMBIOSIS` mid-fight possession, `TIDE_VOYAGER` fishing cultivation, and garden-trap defense (`08D/08E`). 2) Enable autonomous Bounty Hunts, Espionage (with capture/ransom risk!), Exchange Disciples, Black-Market multi-sect bidding, Micro-Location

Phase	Priority	Target Files	Exact Deliverables Across `08A–08F`
			pilgrimages, Immortal Tomb party excavations, Wandering Figure duels/hiring, and 43-Path Rival Sect generational succession (`08A–08F`).